

Understanding the Transition from Memorization to Generalization in Diffusion Models for Cosmology

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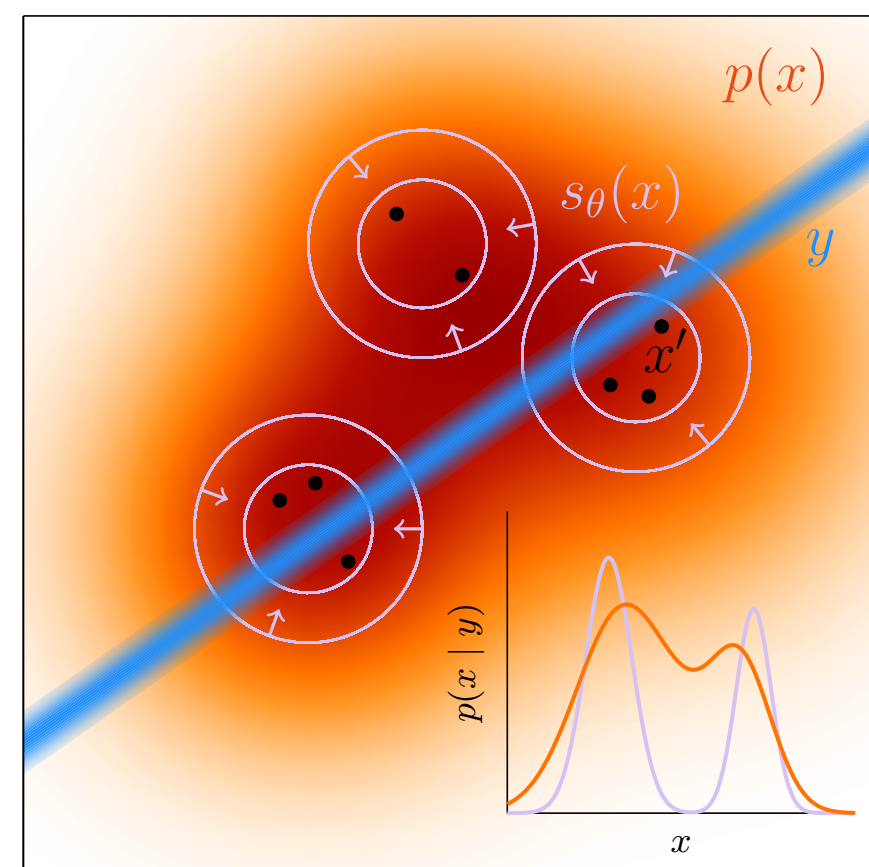
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The question

A trained diffusion model can produce mock HI maps in seconds instead of CPU-hours. That speed is only useful if the maps are genuinely new — a model that memorizes its training set gives a biased prior. We train diffusion models on CAMELS HI maps and ask:



1. When does the model stop **copying** training fields and start **sampling** the underlying distribution?
2. Does conditional generation actually **respond to the requested cosmology** θ ?
3. Are generated fields **statistically faithful** (one-point PDF, power spectrum)?

Toy picture: a memorized prior is a set of islands around the training data. The posterior gets pulled toward those islands even when the likelihood is sharp.

Data & models

- **CAMELS** (IllustrisTNG) LH set: 1000 simulations $\rightarrow$ 128 slices each $\rightarrow$ 128,000 candidate 128×128 HI maps; 32 held out.
- Six-parameter conditioning: $\theta = (\Omega_m, \sigma_8, A_{SN1}, A_{AGN1}, A_{SN2}, A_{AGN2})$.
- **Diffusion objective.** The model learns to undo noise: corrupt a real map, train a UNet to predict the velocity target at every noise level; the cosmology enters as conditioning.

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_0, \epsilon, t} \|f_\theta(x_t, t, \theta) - \text{target}\|_2^2.$$

Here the config uses `v_prediction: target = v_t = \sqrt{\alpha_t} \epsilon - \sqrt{1 - \alpha_t} x_0`. Ho et al. 2020 (DDPM).

- **Sampling:** 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep:** $N_{2D} = 2^6 \rightarrow 2^{15}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

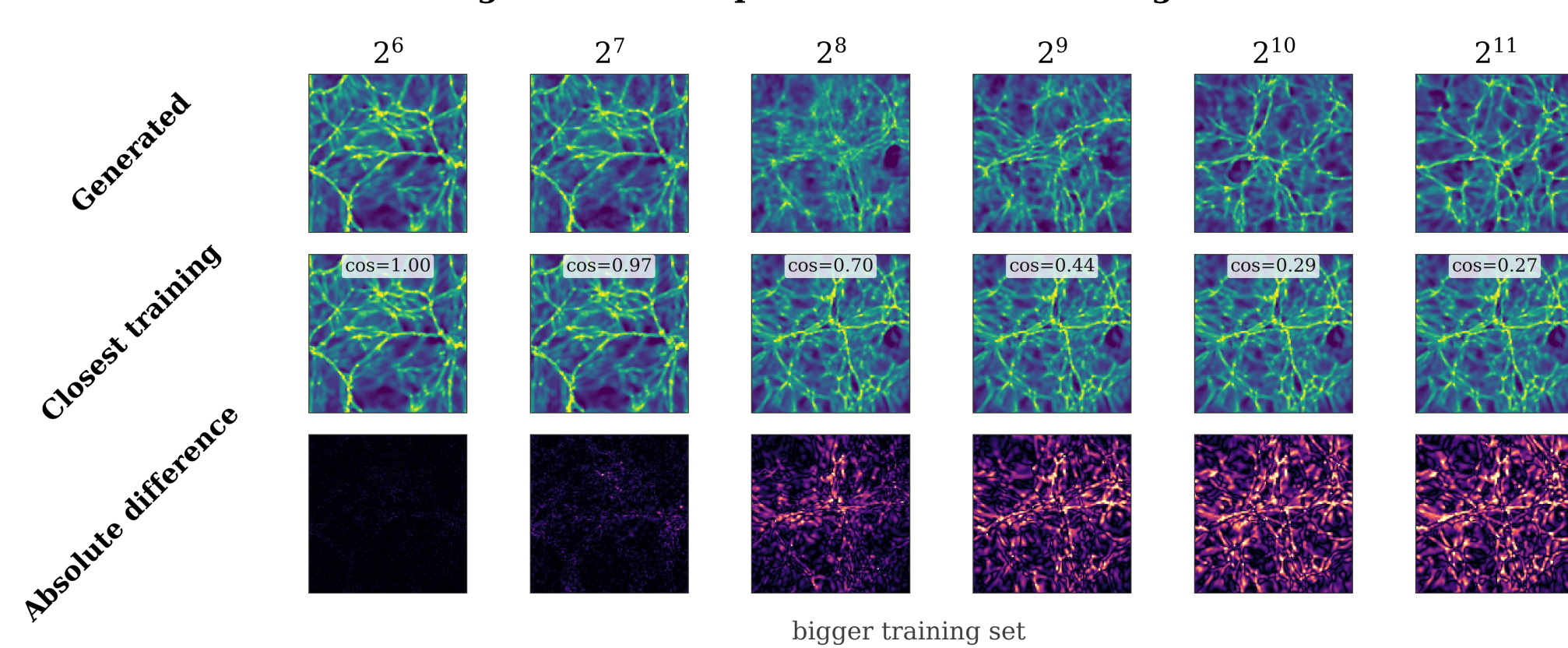
Detecting memorization

For each generated sample x_j , find its nearest training field in pixel, PCA, and SSCD embedding space:

$$s_j = \max_i \text{sim}(x_j, y_i), \quad \text{GL}(\tau) = 1 - \Pr[s_j > \tau]$$

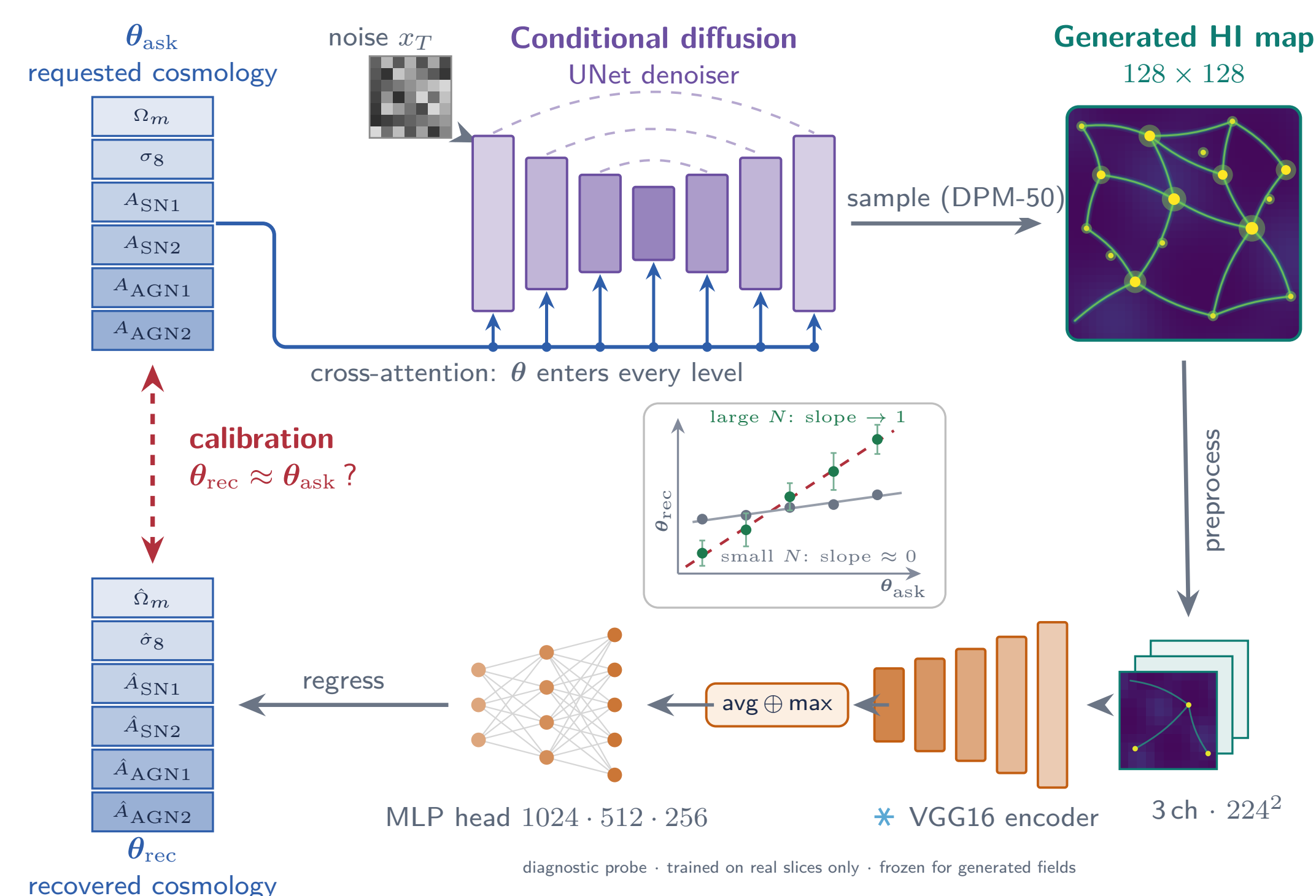
Here τ is set from train-vs-train nearest-neighbor similarity (95th percentile). Independent models should reproduce the same distribution, not the same training images.

UNet-64: generated samples and nearest training slices



Calibration: does the generator obey θ ?

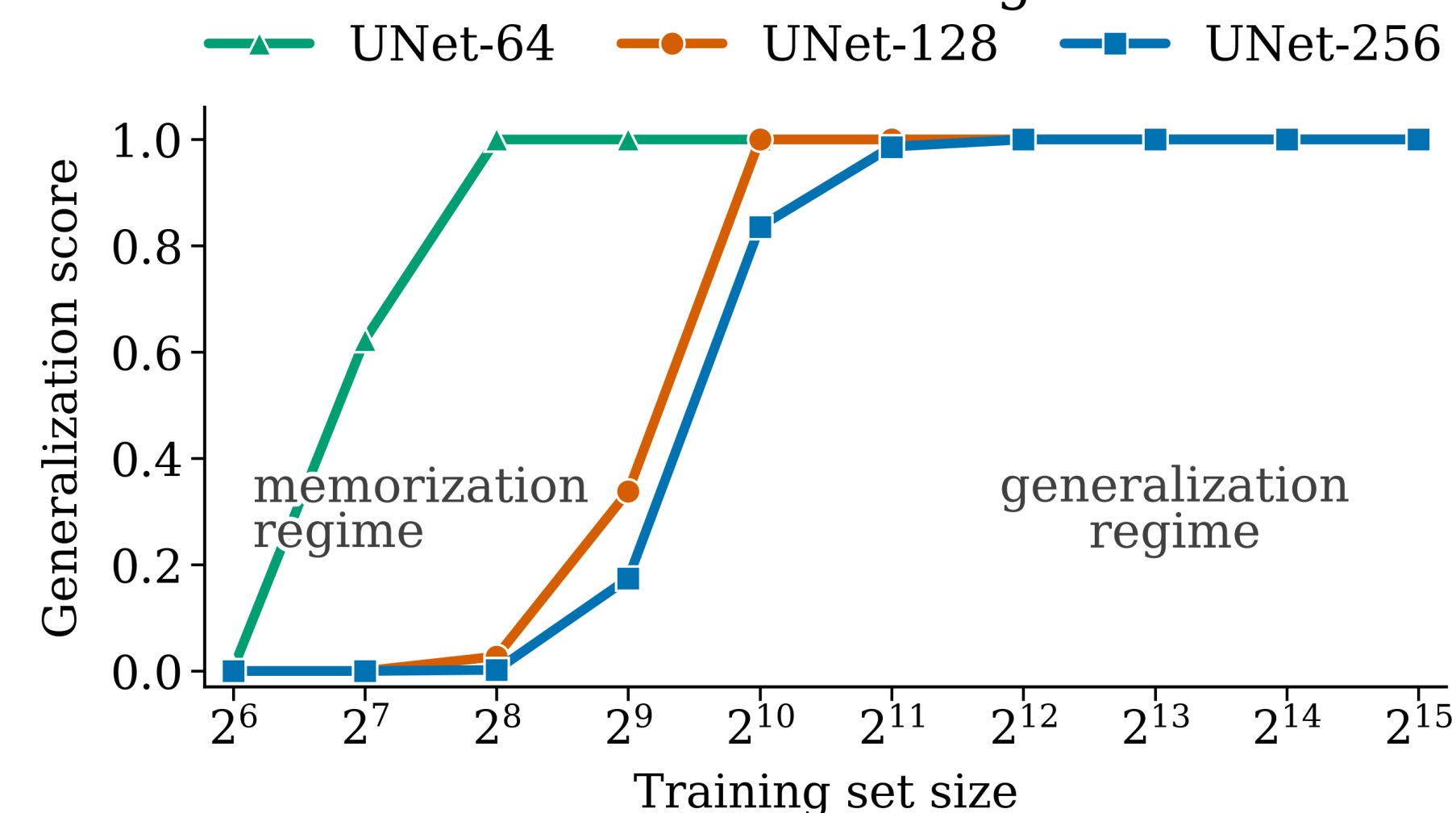
Ask the conditional model for **held-out samples**, then read the cosmology back from its generated maps with a **frozen encoder**. If the conditioning is used, changing θ_{ask} should move the recovered θ_{rec} rather than leaving it near the training-set average.



Encoder setup: VGG16 is frozen (ImageNet); only the regression head is trained on *real* slices from non-held-out simulations. Held-out test simulations are used only for encoder validation and generator calibration.

Memorization $\rightarrow$ generalization transition

Generated fields become less training-set-like with more data

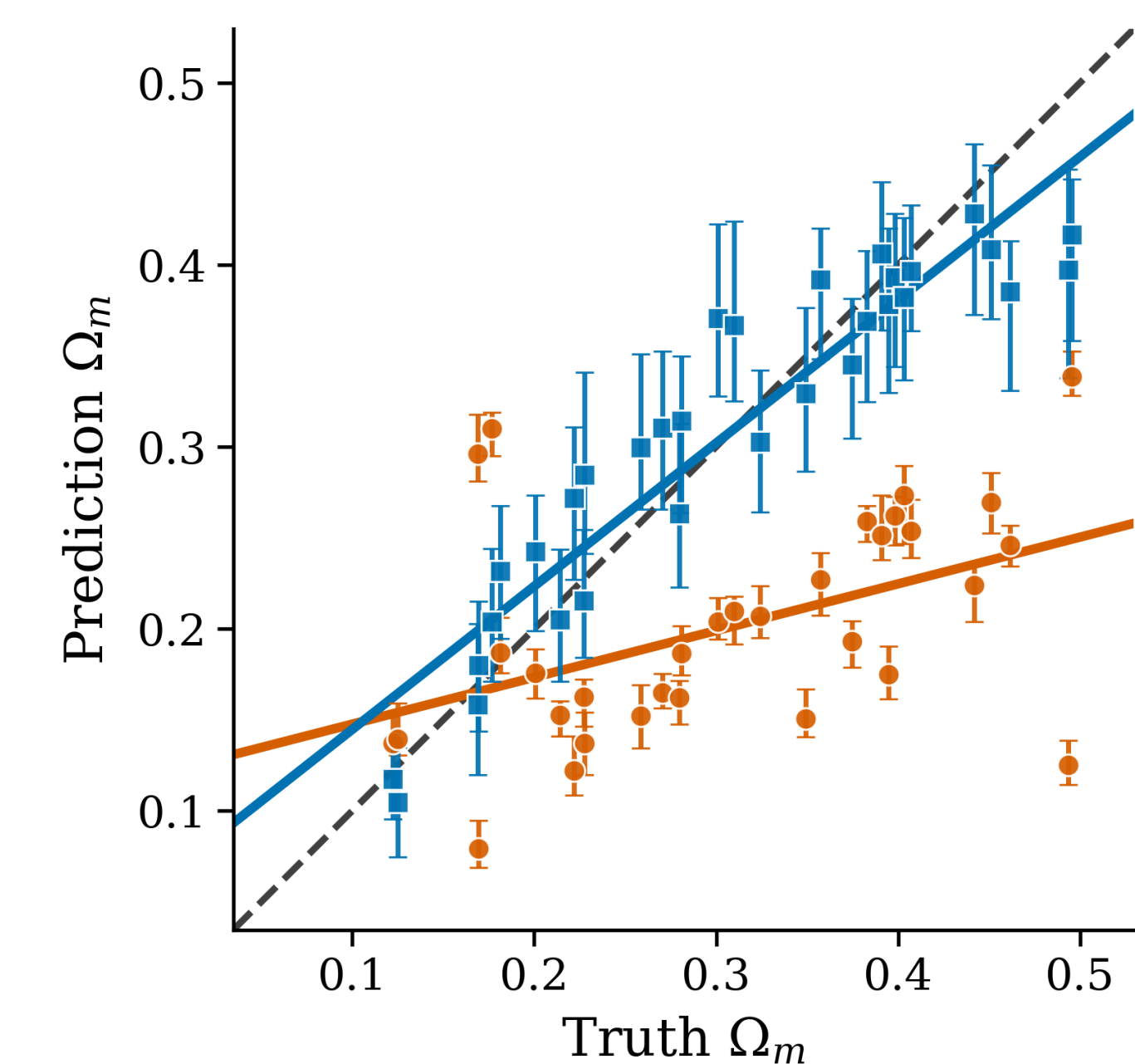


- Score uses a **q95-calibrated threshold** ($\tau = 95\text{th pct train-train NN similarity}$): higher means generated fields are less training-set-like in PCA space.
- **Small N_{2D} :** generated maps remain close to training fields.
- **Larger N_{2D} :** UNet-64/128/256 move toward novel samples; the wider UNet-256 transition is shifted to larger data.

Calibration results

Generated HI fields recover Ω_m

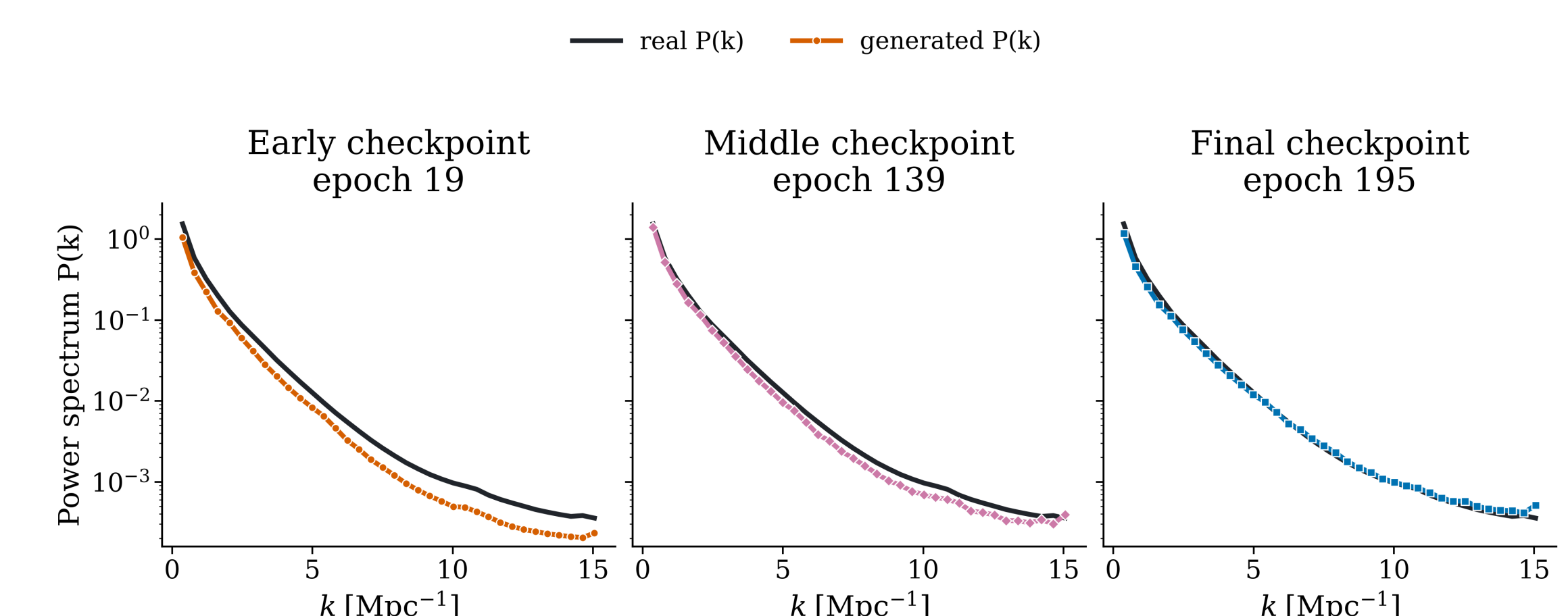
— ideal recovery ● mem. regime ($N_{2D} = 2^7$), slope=0.26 ■ gen. regime ($N_{2D} = 2^{14}$), slope=0.79



Main result: error bars show 16–84% seed scatter at fixed θ_{ask} . In the high-data regime they bracket the requested Ω_m ; under memorization they can be small but centered on the wrong value. Slopes (0.26 vs 0.79) are secondary summaries; Ω_m is shown because the held-out VGG encoder is strongest there ($R^2 = 0.91$).

Physical fidelity

Generated power spectra approach the real reference during training



Across a single run, generated power spectra move toward the real reference from early to final checkpoints. Early checkpoints suppress power over many k bins; by the final checkpoint the generated $P(k)$ follows the real shape much more closely.

Takeaways

1. **Memorization is detectable.** Nearest-neighbor and reproducibility tests locate a data-size transition in CAMELS HI maps.
2. **Conditioning is recoverable.** High-data generated fields track θ_{ask} much better than memorization-regime fields.
3. **Fidelity requires generalization.** Useful emulators must be novel, statistically faithful, and calibrated before inference use.